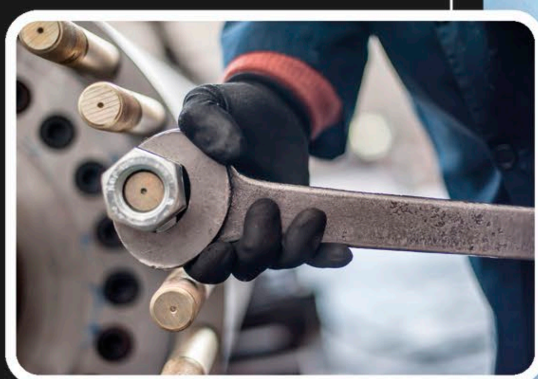
LOAD LIFTERS

CRANES

Building bridges and skyscrapers means heavy loads need to be lifted. This can be done by large machines called cranes, which make use of levers and pulleys. Along with screws and wheels-and-axles, pulleys and levers are all examples of simple machines—devices that can change the direction of forces and convert large forces into small ones or small forces into large ones. In nearly all cranes, a system of pulleys—wheels with rope passing over them—is mainly responsible for lifting the load.

WRENCHES

A wrench, which is used to turn nuts and bolts, is a type of simple machine called a lever. Levers turn about a fixed point called a fulcrum (here at the head of the wrench). Pushing on the wrench away from the fulcrum and farther down the wrench's long handle increases the force at the fulcrum.



PULLEYS

A pulley makes it easier to lift a load. With a single pulley, you can lift a load upward by pulling downward on a rope instead—an easier action. When two or more pulleys are combined by connecting them with a single rope—also called a block and tackle—a much greater load can be lifted than the force with which the rope is pulled.